METRIC

MIL-F-71007 (AR)
15 February 1991

MILITARY SPECIFICATION

Fiberscope, Explosive Ordnance Disposal (EOD)

This specification is approved for use by the US Army Armaments Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and tests for the Fiberscope, Explosive Ordnance Disposal (EOD), for use by EOD technicians.

2. APPLICABLE DOCUMENTS:

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-7298 Manuals, Commercial Off-the-shelf

MIL-A-48078 Ammunition, Standard Quality Assurance Provisions, General Specification for.

STANDARDS

MILITARY

MIL-STD-210 Climatic Information to Determine Design and Test Requirements for Military Systems and Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6650

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(Unless otherwise indicated, copies of federal and military specification, standards, and handbooks are available from the Standardization Documents Order Desk , Bldg. 4D, 700 Robins Avenue, Philadelphia. PA 19111-5094.)

2.1 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item Definition. The EOD Fiberscope shall be a mature design, currently available off the shelf device which has been in production for at least one year. It shall be a portable, flexible, fiber optic device that is capable of illuminating an objective area and capturing and transmitting a visual image from the objective end to the eyepiece. It shall have the features as specified in 3.1.1 through 3.1.10.10.

3.1.1 Operation. The fiberscope shall be capable of being safely carried and operated by one person.

3.1.2 Set up time. The fiberscope shall be capable of being unpacked from its case and made operational within five (5) minutes. The device, its associated components, and its collective carrying case shall not require the use of tools to set up, use, or transport.

3.1.3 Safety. The design of the fiberscope shall not induce human error and shall be safe to transport, set up, and use. The design of the fiberscope, its associated components, and carrying case shall comply with applicable health, safety, and human engineering design requirements and shall not present any uncontrolled health-safety hazards to personnel throughout the life cycle of the item.

3.1.4 Compatibility. The device shall be compatible with, and its use shall not be hindered by, the wearing of cold weather equipment or NBC (MOPP IV) protective clothing and equipment. The use of this device while wearing Arctic mittens is not required. However, use while wearing other related protective handwear is required.

3.1.5 Static electricity. The fiberscope shall not act as source of static electricity.

3.1.6 Articulation. The fiberscope shall have an articulation control to bend the objective end of the shaft by 90 degrees up and down or side to side, coplanar within ten degrees.

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3.1.7 Nonconductivity. The surface coating of the fiberscope shaft shall be nonconductive (10,000,000 ohms/centimeter or greater).

3.1.8 Immersible. The fiberscope shaft shall be immersible in water, fuels, and oils without damage. The shaft shall not be adversely affected by chemical decontaminants.

3.1.9 Eyepiece focus. The eyepiece shall have an adjustable focus to allow operation by personnel wearing corrective eyeglasses or using chemical or explosive protective equipment.

3.1.10 Fiberscope system components. The fiberscope system shall consist of the following items:

3.1.10.1 Fiberscope. Fiberscope, consisting of fiber optic image and light transmission bundles in a flexible shaft, objective lens, eyepiece, and articulation control.

3.1.10.2 Eyecup. A removable, flexible eyecup shall be provided to maintain proper eye relief and eliminate stray light. It shall be compatible with helmets, protective masks, and other clothing and protective equipment.

3.1.10.3 Illuminator and battery. Illuminator with battery and associated equipment required to provide light to the light transmission bundle.

3.1.10.4 Battery charger. The fiberscope system shall include the equipment to recharge the illuminator battery using 110 volt, 60 cycle and 220 volt, 50 cycle power. Recharge time shall be 12 hours or less.

3.1.10.5 Viewing adapter. Right angle viewing adapter for fiberscope tip.

3.1.10.6 Video camera adapter. Video camera adapter, "C" mount type.

3.1.10.7 Spare battery. Spare illuminator battery.

3.1.10.8 Spare bulbs. Two spare illuminator bulbs.

3.1.10.9 Operation manual. Operation and maintenance manual in accordance with MIL-M-7298 (see 6.3).

3.1.10.10 Carrying case. Carrying case that holds all components and provides protection for components during air and vehicle transportation and during storage and shipment. The case shall permit rapid access to the device and its components and shall provide for hand-carry. The storage case shall protect the fiberscope and accessories from chemical contamination and shall be capable of being decontaminated.

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3.2 Characteristics

3.2.1 Performance

3.2.1.1 Field of view. The fiberscope shall provide an objective field of view between 50 and 70 degrees, when tested as specified in 4.4.1.

3.2.1.2 Depth of field. The fiberscope shall provide a clear, focused image at objective to target distances between 25 and 100 mm, when tested as specified in 4.4.2.

3.2.1.3 Resolution. The fiberscope shall provide the operator an initial image with resolution of 1.0 line pairs per millimeter (Lp/mm) or greater, using the battery powered illuminator, measured on a U.S. Air Force 1951 Resolution Chart at a range of 50 mm, when tested as specified in 4.4.3.

3.2.1.4 Operational mission time. The fiberscope illuminator battery shall have a life of 2 hours or greater without recharging. At 2 hours, resolution shall not be less than 0.7 Lp/mm, measured on a U.S. Air Force 1951 Resolution Chart at a range of 50 mm, when tested as specified in 4.5.4.

3.2.2 Physical Characteristics

3.2.2.1 Operational weight. The fiberscope shall have an operational weight of 6.8 kilograms (15 pounds) or less, including fiberscope, illuminator, and battery.

3.2.2.2 Total system weight. The fiberscope system shall weigh no more than 18.1 kilograms (40 pounds), including its shipping case and all cited accessories.

3.2.2.3 Working diameter. The fiberscope shaft shall have a maximum diameter of 13 millimeters or less.

3.2.2.4 Working length. The fiberscope shall have a working (flexible shaft) length between 1.4 and 1.8 meters.

3.2.2.5 Bend radius. The fiberscope flexible shaft shall have a bending radius of 50 millimeters or less, without damage to the fiber bundle.

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3.2.3 Maintainability

3.2.3.1 Preventative maintenance. Routine preventative maintenance shall be accomplished at the operator level.

3.2.3.2 Parts replacement. The design of the illuminator shall allow for field replacement of illuminator bulbs, fuses, and batteries.

3.2.4 Environmental Conditions

3.2.4.1 Operational temperatures. The fiberscope shall be capable of being operated and stored in basic, hot, and cold climatic ranges in accordance with MIL-STD-210 (-12 to 52 degrees Centigrade) (10 to 125 degrees Fahrenheit).

3.3 Design and construction

3.3.1 Design. The device and its components shall be free of small, loose parts that could be easily lost or misplaced. To the maximum extent possible, all components shall be integral or captively held to the device. The device, its components, and the carrying case shall be free of sharp edges and other protrusions which could cause injury.

3.3.2 Workmanship. All parts shall be fabricated and finished in a thorough, workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease, oil, corrosion products and other foreign matter. Cleaning methods used shall not be injurious to any part or assembly, nor shall they be contaminated by the cleaning agent.

3.3.3 Fiber defects. The fiberscope shall have no visible voids in the viewing area. The fiberscope shall have less than 0.05 percent broken optical fibers and voids.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor shall be responsible for the performance of all inspection requirements as specified in the contract. Except as otherwise specified in the contract of purchase order, the contractor may use his own or any other facility suitable for the performance of the inspection requirements specified herein. The government reserves the right to perform any of the inspections (tests) set forth in this specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item.

a. First article inspection.

b. Item acceptance testing and inspection.

4.2.1 Characteristics. The fiberscope shall meet the requirements specified in 3.2.

4.2.2 Functional test. The fiberscope shall be subjected to a normal test to verify requirements of 3.1.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with Table I.

4.3.2 Inspection to be performed. The first article parts submitted in accordance with 4.3.1, will be subjected by the Government to any or all the requirements of the applicable drawings, and the inspections and tests of Table I.

4.3.3 Rejection. See MIL-A-48078. The first article sample shall be rejected if any assembly or subassembly fails to comply with any of the examinations ot tests specified in this specification, or any of the requirements of the applicable drawing.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-F-71007 (AR)

PARAGRAPH	TITLE Explosive Ordnance Disposal, Fiberscope	SHEET 1 OF 1	DRAWING NUMBER NEXT HIGHER ASSEMBLY	
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Set Up Time	1, 0-1	3.1.2	Gage
	Articulation	1, 0-1	3.1.5	Manual
	Eyepiece Focus	1, 0-1	3.1.8	Gage/ Visual
	Fiberscope, components	1, 0-1	3.1.9	Visual
	Performance		3.2	
	Field of View	1, 0-1	3.2.1.1	Gage/ Visual
	Depth of Field	1, 0-1	3.2.1.2	Gage/ Visual
	Resolution	1, 0-1	3.2.1.3	Gage/ Visual
	Operational Mission Time	1, 0-1	3.2.1.4	Gage

NOTES

AMSMC Form 1570a-E, 1 Jul 89

Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Examination and test.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraph. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein.

b. Examination. See MIL-A-48078 and Table II herein.

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TABLE II. ITEM ACCEPTANC :STING AND INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-71007 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4	Explosive Ordnance Disposal, Fiberscope		NEXT HIGHER ASSEMBLY ----

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Set Up Time	100%, 0-1	3.1.2	Gage
	Articulation	100%, 0-1	3.1.5	Manual
	Eyepiece Focus	100%, 0-1	3.1.8	Gage/ Visual
	Fiberscope, components	100%, 0-1	3.1.9	Visual
	Performance		3.2	
	Field of View	100%, 0-1	3.2.1.1	Gage/ Visual
	Depth of Field	100%, 0-1	3.2.1.2	Gage/ Visual
	Resolution	100%, 0-1	3.2.1.3	Gage/ Visual

NOTES

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4.5 Methods of inspection. Testing and inspection shall be performed at 100% of all items in accordance with the procedures specified in Table II and optical performance tests herein.

4.5.1 Field of view. A Government approved target representing the requirement as stated in 3.2.1.1 shall be developed by the contractor. This target shall be viewed at the appropriate distance as relates to the target dimensions perpendicular to its center. The fiberscope shall, when viewed with the unaided eye, and when properly focused, display the entire field.

4.5.2 Depth of field. A government approved fixture shall be used to measure the depth of field. While viewing through the fiberscope with the unaided eye, there shall be a clear focused image at the target distances as specified in paragraph 3.2.1.2

4.5.3 Resolution. The resolution target used in this test shall conform to the U.S. Air Force 1951 resolution chart. Illumination of the resolving power chart shall be by the fiberscope. An auxiliary telescope of at least three power (3X) shall be used at the fiberscope eyepiece to obtain a clear, sharp image of the target. Alignment of the fiberscope and auxiliary telescope shall be such that a true image is observed. The line pair group which represents the resolution requirement and the distance from the objective head as defined in 3.2.1.3 shall be clearly marked or referenced. To be acceptable, the correct line group shall be clearly discernible when viewing through the auxiliary telescope.

4.5.4 Operational mission time. The fiberscope illuminator shall operated for two (2) hours minimum. At that time the resolution shall meet the requirements of 3.2.1.4

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with Special Packaging Instructions as specified in the contract by the procuring activity.

6. NOTES

(This section contains information of a general or explanatory nature that may helpful, but is not mandatory).

6.1 Intended use. This fiberscope will be used by U.S. Army EOD technicians to accomplish search and inspection duties involving conventional ordnance and improvised explosive devices.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.

6.3 Technical manual acquisition. This specification must be listed on the Contract Data Requirements List (DD Form 1423) in order to acquire the technical manual described by this specification (3.1.10.9), except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for sample for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Reference documents. The following documents are listed for information only.

MIL-STD-109 Quality Assurance Terms and Definitions (for reference only)
MIL-STD-1472 Human Engineering Design Criteria for Military Systems, Equipment, and Facilities (for reference only)

6.6 Drawings. Drawings shall be contractor generated.

6.7 Subject term (keyword listing):
Fiberscope
Eyepiece focus

Custodian:
Army-AR

Preparing activity
Army-AR

(Project 6650-A186)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-71007 (AR)	15 February 1991

3. DOCUMENT TITLE
FIBERSCOPE, EXPLOSIVE ORDNANCE DISPOSAL (EOD)

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (*Include Area Code*)	
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c. ADDRESS (*Include Zip Code*) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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